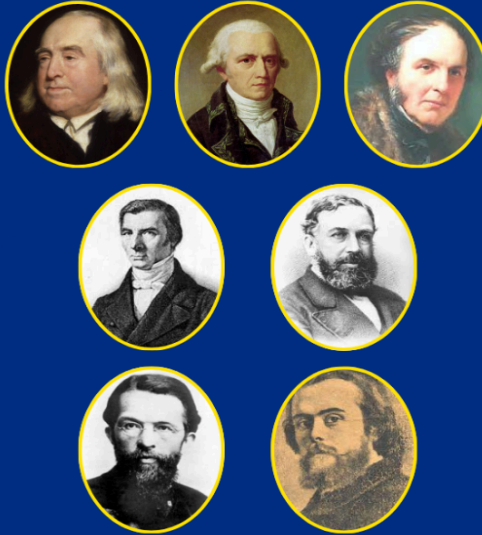




Bentham to Walras: Economic Ideas and Beyond



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Submitted to:

Kawsar Hossain Sojib

Lecturer

Department of Economics

Jahangirnagar University

Savar, Dhaka

Submitted by:

Group Name: **THE INVINCIBLES**

Group Members:

- Irteza Alam Oishe (445)
- Moni Nasrin (448)
- Mst. Ashrafi Siddika (449)
- Sadia Afrin (455)
- Umme Habiba (465)
- Anita Azad Riya (471)
- Afif Ibn Munir (490)
- Md. Motiur Rahman (491)
- Wasif Fuad (505)

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Table of Contents

Introduction.....	4
Jeremy Bentham.....	5
Jean-Baptiste Say.....	9
Nassau Senior.....	14
Fedric Bastiat.....	17
William Stanley Jevons.....	21
Carl Menger.....	24
Leon Walras.....	29
References.....	31

Introduction

One of the branches of the classical school of economics is “Neoclassical economics”. This group features some of the remarkable economists in the history of economics. For example: Bentham, Say, Jevons, Menger etc. Neoclassical economists came up with the idea of utility theory and added further to the discussions of classical economists with the likes of Adam Smith, Ricardo etc. Furthermore, they also contributed to the marginal idea which is part and parcel of modern economics. Thus, it is important to know what their contributions were. This writing makes an attempt to shed light on some of the key ideas of prominent Neoclassical economists.

Jeremy Bentham

Jeremy Bentham was an influential English philosopher, jurist, and social reformer, born on February 15, 1748, in London, England. He is best known for his development of utilitarianism, a moral theory that advocates actions that maximize happiness and minimize suffering for the greatest number of people. Bentham believed in the principle of the greatest happiness of the greatest number, emphasizing the importance of rational calculation in ethics and law. Throughout his life, Bentham advocated for various social reforms, including prison reform, animal rights, freedom of expression, and the separation of church and state. He also made significant contributions to legal theory, particularly in the areas of legislation and jurisprudence. Bentham's ideas greatly influenced political thought and had a lasting impact on legal and ethical discourse. He died on June 6, 1832, but his legacy continues to shape contemporary debates in philosophy, law, and politics.

Utilitarianism received its most distinctive, classical formulation in the writings of Jeremy Bentham (1748-1882) whose work, that was the most powerful influence on nineteenth-century economic theory, "An Introduction to the Principles of Morals and legislation" published in 1780. Although it does not deal directly with economic theory, it contains an elaborate statement of the utilitarian social philosophy which has become the philosophical basis of neoclassical economies during the last several decades of the nineteenth century.

Bentham started his book with the statement that, "Nature has placed mankind under the governance of two sovereign masters, pain and Pleasure." And he began with the assertion that all human motivation, in all times and all places, can be reduced to a single principle: the desire to maximize one's utility. He also stated utility as: "By utility is meant that property in any object, whereby it tends to produce benefit, advantage, pleasure, good, or happiness, to prevent the happening of mischief, pain, evil, or unhappiness to the party whose interest is considered".

So Bentham believed that all human motivation were merely manifestations of the desire for Pleasure and the avoidance of Pain. Because pain was merely negative pleasure, Bentham's principle of utility could be restated as "All human activity springs from the desire to maximize pleasure"

By reducing all human motives to a single principle, Bentham believed that he had found the key to the construction of a science of human welfare or happiness that could be stated mathematically and might someday be worked out with the same numerical exactitude as the

science of physics." Pleasures. and the avoidance of pains are... ends", he argue, which can be numerically quantified such that we can "understand their value".

He suggested a possible method for quantifying Pleasures:

"To a number of persons, with reference to each of whom the value of a pleasure or pain is considered, it will be greater or less, according to seven circumstances:

1. Its intensity.
2. Its duration.
3. Its certainty or uncertainty.
4. Its propinquity or remoteness.
5. Its fecundity.
6. Its purity.
7. Its extent."

He then went into a discussion of the specific ways in which these circumstances might suggest how one could rationally calculate the values of pleasures and pains. Bentham not only conceived of human beings as calculating maximizers of pleasure, he also saw them as being fundamentally individualistic. "In the general tenor of life," he wrote, "in every human breast, self-regarding interest is predominant over all other interests put together. Self-preference has a place everywhere". He believed, People were also essentially lazy. Any kind of exertion on work was viewed as painful, and, therefore, work would never be undertaken without the promise of greater pleasure or the avoidance of greater pain.

Each of these ideas of Bentham was to become more important in the subsequent development of the utility theory of value. Bentham believed: "All value is founded on utility... where there is no use, there cannot be any value".

Smith and Ricardo would have agreed that use value was a necessary condition for exchange value. But, as Ricardo insisted, when one sees value as created by labor, then an increase in the productivity of labor lowers the value of a commodity while increasing the general wealth, when Bentham stated that "it is by the degree of that value that Wealth is measured", he was speaking from the utility theory perspective, in which an increase in utility increases a commodity's value and hence increases its owner's wealth.

Thus, Bentham not only formulated the philosophical foundation of the later tradition of neoclassical economics but also came very close to developing a theory of the relationship between marginal utility and price. The development of his ideas also foreshadowed an important split in the orthodox utility approach to economics.

The main basis of Bentham's hedonism is self-hedonism. So the question is, if people want their happiness, why would they want the happiness of others? That is, how to go from an individual's happiness to everyone's happiness?

To explain it Bentham talked about four external sanctions. They are:

1. Social Sanction
2. Political Sanction
3. Religious Sanction
4. Physical or Natural Sanction

Bentham as a Social Reformer

In Bentham's earlier writings, he accepted Smith's argument that a competitive free market would allocate productive resources to those industries in which they would be the most productive. He therefore believed that government interference in the free market might very likely diminish the level of production.

Bentham also accepted, during his early writings, that aggregate supply would always equal aggregate demand in a free market. In such a market there need never be a depression or any involuntary unemployment because any savings was automatically converted into more capital with which to employ more labor.

However, by 1801, Bentham's opinions about government intervention in the economy had undergone a change and he stated, "The interference of government, and often as... (it results in) the smallest... advantage... is an event I witness with... satisfaction."

This change of opinion was prompted by two principal concerns, each of which was later to become an important argument against complete laissez faire.

- First, Bentham, like Malthus, came to see that savings might not be matched by new investment. In this event, production would diminish, unemployment would be created,

and the free market would not be functioning in the best interests of the public. He stated that an increase in the habit of frugality would diminish the mass of expenditure on consumption and disrupt the money supply. In that case, saving would lead to decreased prices and production; profits would decline and so would investment. In such a case, Bentham argued, if the government increased the amount of money in circulation, "then the money introduced • becomes a source of increasing Wealth"

- The second reason for government interference in the market was to lessen the social effects of great inequalities of wealth and income. Bentham believed that a person's capacity to get enjoyment from money declined as he got more money. In modern terminology, he believed in a diminishing marginal utility of money. Therefore, all other things being equal, a government measure that redistributed money from the rich to the poor would increase the total of society's aggregate utility. [Government had to be a benevolent, impartial institution, equally interested in every individual's welfare]

Thus, over Bentham's lifetime, there was a split or antagonism between his earlier extreme laissez-faire attitude and his later reformist attitude. This same split was to be reflected later in the neoclassical tradition, which was constructed on the foundation of Bentham's utilitarian philosophy.

Jean-Baptiste Say

Jean-Baptiste Say (5 January 1767 – 15 November 1832) was a liberal French economist and businessman who argued in favor of competition, free trade and lifting restraints on business. He is best known for Say's law—also known as the law of markets—which he popularized.

J. B. Say, was originally a follower of Adam Smith's ideas. His thoughts were organized around Smith's ideas and he tried to improve upon them. He believed that there were some small mistakes in Smith's ideas, so he worked to correct those mistakes by using Smith's overall ideas. That is, instead of creating something entirely new, Say wanted to perfect small mistakes based on what Smith had started.

But, as it turns out, these small changes completely abandon Smith's ideas and lead to something completely new. This led to a completely different way of thinking about economics than Smith originally intended.

One of Smith's mistakes is that Smith believed that only human labor (people working) can create value, and based on this, he provided the Labor Theory of Value. Say rejected Smith's labor theory of value. According to him, this is not entirely true as there are other things besides just human labor that can add value to goods and services.

According to Say, the value or price of something depends on how useful it is. For example, a pencil may not cost much because it is easy to find and many people have it, but if you are stranded on an island without any writing equipment, a pencil will be very valuable because it will help you communicate and keep track. So basically, the more useful a product is to someone, the more they will be willing to pay for it.

Thus, he rejected labor as the source of value and added that utility creates value. He said that, "the utility of things is the ground-work of their value, and their value constitutes wealth. Exchangeable value, or price, is an index of the recognized utility of a thing."

In rejecting the notion that labor was the source of value and insisting that only utility created values, Say not only departed sharply from the ideas of Smith and Ricardo, but also he placed the utility approach in the context of a methodological approach and a social philosophy.

According to Smith and Ricardo, the income of labor differs from the income of ownership of the factors of production, and in looking for the source of this difference, they conclude that class conflict characterized capitalism. But, after that, when they reverted to the exchange or utility approach to economic theory, they were led to the conclusion that free market capitalism was inherently a system of social harmony.

Here lies the dilemma.

Say resolved this dilemma by rejecting the labor theory of value approach to economic theory.

In his utility framework

1. He rejected the theoretical distinction between income of different social classes.
2. Rather than seeing the production process as a series of human efforts to transform various natural raw materials into usable products, he emphasized the existence of various 'productive agencies' that combine to produce goods.
3. These productive agencies were ultimately producing as "utility," and each agency was coequally responsible for the production of the utility.
4. These productive agencies included "human industry, with the aid of capital and of natural agents and properties," altogether created "every kind of utility, which works as the primary source of value. In other words, there was no qualitative difference, in the creation of utility, between the exertion of human labor and the ownership of capital, land, and other property.

In his utility framework, Say wanted to show that owners of the means of production received their incomes from similar sacrifices. Smith and Ricardo had argued that commodity prices reflected the rate of wages and the rate of profits and that these rates were determined by other social and technical considerations, where as, Say argued that wage and profit rates were determined by the relative contributions to utility creation made by labor and capital.

Within Say's utility approach to value and distribution theory, all notions of class conflict disappeared. A central purpose of Say's Treatise was to demonstrate that social harmony was the natural result of a capitalist economy, not class conflict.

Say's Law of Markets:

Say's Law of Markets states that the supply of a good or service creates demand for that good or service, that is - **supply creates its own demand**. And that economic agents must first engage in production before they can demand goods and services in the market.

According to the Law of Markets, a person's ability to demand goods and services is a direct result of production activities that they've undertaken. They earn an income either through the production and sale of physical assets or by supplying labor to capital owners. The productivity of the economic agent will influence the level of income they receive, which will, in turn, influence what goods and services they demand and how much of each good or service they

demand. By choosing to produce a certain good, they are indirectly creating the demand for those goods and services that they intend to buy with their earned income. Conversely, if one did not demand anything, they would not want to work. Thus, Say's Law describes the unfolding market process, stating that "all purchasers must first be producers, as only production can generate the power to purchase."

According to Say, there may be a temporary surplus of some products, but this will result in the market not achieving equilibrium.

In this case, there will be a surplus of those products that were priced too high and simultaneously a shortage of those products that were priced too low. Surpluses and deficits will just cancel out as a whole. Moreover, capitalists who sell goods at a too low price will make low profits, and those who sell at a too high price will make high profits. Then, the search for maximum profits will cause capitalists to leave low-cost industries and enter high-cost industries. This migration of capitalists will have two effects.

First, it will change prices, raising the prices of low-priced goods and lowering the prices of high-priced goods.

Second, it would reduce the production of those goods for which there was a surplus and increase the production of those goods for which there was a shortage.

Thus, throughout the process, aggregate supply will always equal aggregate demand. And demand and supply will be equal for each product as well. Hence, there can never be a surplus of all products in the market. Moreover, temporary shortages and surpluses between different products will be eliminated automatically by free, competitive markets.

These conclusions form Say's Law of Markets and are still accepted by many economists and policy makers today.

Propositions and Implications of the Law:

Say's propositions and its implications shows us the proper scenario of the marketplace regulation. Also, We can say these are the assumptions behind the law. These are given below:

1. Full Employment in the Economy: The regulation is primarily based totally at the proposition that there's complete employment within the economy. Production keeps growing till the extent of complete employment is reached. Under one of these situations, the production might be maximum.

2. Proper Utilization of Resources: If there's complete employment within the economy, idle assets might be used properly to generate extra income.

3. Perfect Competition: The regulation is primarily based totally at the proposition of best opposition in labor and product markets.

4. Automatic Adjustment Mechanism: The regulation is primarily based totally in this proposition that there's an automated and self-adjusting mechanism in the markets. Disequilibrium in any marketplace is a temporary situation.

5. Money is neutral: It is based on the proposition of a barter system. So, it assumes that the position of money is neutral. Money has no effect on the production process.

6. Laissez Faire Policy: The regulation assumes a closed capitalist economic system which follows the coverage of laissez-faire.

Criticism:

Violating the assumptions of Say's Law, Keynes has criticized the law. He criticized Say's law of markets on the subsequent grounds:

1. Supply does not create its Demand: The proposition of the law - Supply Creates Its own demand - is not applicable to modern economies where demand does not increase as much as production increases. It is also not possible to consume only those goods which are produced within the economy.

2. Self-adjustment is not Possible: According to Say's law, full-employment is maintained by an automatic and self-adjustment mechanism in the long run. But Keynes had no patience to wait for the long period for he believed that "In the long-run we are all dead."

3. Money is not Neutral: Say's law of markets is based on a barter system and ignores the role of money in the system. But Keynes has given importance to money. He regards money as a medium of exchange. Money is held for income and business motives.

4. Over Production is Possible: According to Keynes, all income accruing to factors of production is not spent but some fraction out of it is saved which is not automatically invested. So, over-production may be possible.

5. Underemployment Situation: Keynes regards full employment as a special case because there is underemployment in capitalist economies. Millions of workers are prepared to work at the current wage rate, and even below it, but they do not find work.

6. State Intervention: Say's law is based on the existence of laissez-faire policy. But Keynes has highlighted the need for state intervention in the case of general overproduction and mass unemployment.

Nassau Senior

Nassau Senior (September 1790-June 1864), who was an English lawyer, played a significant role in the early stages of modern neoclassical economics, similar to Bentham and Say. Like Say, he took concepts from earlier classical economists, adjusted some, and introduced his own ideas to create a consistent theoretical justification of the status quo of nineteenth-century capitalism. His ideas about the appropriate methodology of economic theory, the place of utility in explaining value, and the moral and intellectual justification of profit and rent constitute the most important areas in which he influenced the later neoclassical tradition. Senior had a strong interest in social, economic, and political issues.

In 1825, he was appointed to the first chair of political economy at Oxford University. The social and economic problems that most consistently commanded his attention were -

- The general condition of the working class and
- The causes and consequences of poverty.

Senior's ideas about the working class and poverty underwent a dramatic change in 1830, and, in the period after 1830, he produced the ideas that were to have the most influence both intellectually and politically.

Prior to 1830, Senior was a politically conservative man with a keen sympathy and benevolent concern for the poverty of the working class. His 'Introductory Lecture on Political Economy' was published in 1826, and his 'Two Lectures on Population' was published in 1828. In these early works, he showed optimism about the future of the working class. He did not believe that Malthus's population theory legitimately led one to the conclusion that working people would always be at a subsistence level. He believed, rather, that increases in productivity could be accompanied by improvements in the moral character of workers, and, consequently, the standard of living of most workers would rise. He actively supported efforts that he believed would uplift the intellectual and moral status of the poor, and he saw moral education as the only hope for eliminating poverty. However, his views were to change in 1830.

Between 1829 and 1842, England experienced a long series of labor difficulties. Industrialization had reduced the English working class to an almost subhuman level of exploitation and degradation. In the 1820s and 1830s, the working class fought back. After 1829, there were many massive efforts to organize labor, which frequently met with harsh repression. The consequence was widespread strikes, riots, and industrial sabotage, all of which profoundly frightened Senior. Senior became convinced that the poor laws and the government's dole to the poor and the unemployed were the principal causes of poverty and a great threat to the very existence of English capitalism.

In 1830, he published 'Three Lectures on the Rate of Wages', written in the early part of that year. In this preface, he argued that capitalists had a "fund for the maintenance of laborers" (this notion came to be known in economic literature as the "wages fund doctrine"). He

asserted that the size of this fund was determined solely by labor's productivity. Therefore, improving the living standard of workers required either an increase in their productivity or a decrease in the number of workers among whom the wages fund was divided. There were, he stated, two ways of increasing labor's productivity:

- First, the removal of all restrictions on free commerce and accumulation of capital, and
- Second, the abolition of the poor laws, which had "made wages not a matter of contract between the master and the workman, but a right for the one, and a tax on the other."

In the preface, it is clear that Senior was no longer concerned with the misery caused by poverty, but with "the threat of an arrogant laboring class, resorting to strikes, violence, and combinations [unions], a threat to the foundations not merely of wealth but of existence itself."³⁶ The great danger, in Senior's eyes, was that labor unions would fight to maintain and extend the notion that wages should reflect the needs of each worker's family rather than the free play of the forces of supply and demand. The poor laws were based on a system of family allowances for the unemployed and the destitute. Such laws, Senior believed, decreased workers' incentives to work and created the arrogant attitude of workers that their families had a right to exist even if the workers themselves did not or could not find work. This created an "unnatural" relationship between capitalists and workers.

In this period, the ideas of radicals and socialists were spreading fast. Thomas Hodgskin and William Thompson had written books in the 1820s in which they argued that profits were generally unearned income coercively expropriated from workers.³⁹ These ideas were being widely disseminated and discussed. The socialist doctrines of Robert Owen were also rapidly becoming influential. Senior deplored socialist ideas, calling them a mistaken belief that the conditions creating inequality could be remedied. Such a mistaken belief he called "the political economy of the poor," and he argued that these ideas had a natural appeal to the uneducated. Equality can only involve extreme misery, he argued, because "though it is in the power of human institutions to make everybody poor, they cannot make everybody rich; ... they can diffuse misery, but not happiness."

The new poor law reflected the following commission views:

- (1) Workers should accept any job the market offered, regardless of the working conditions or pay involved;
- (2) Any person who would not or could not find work should be given just barely enough to prevent physical starvation; and
- (3) The dole given to such a person should be substantially lower than the lowest wage offered in the market.

It is also very clear that Senior wanted his theory to be seriously considered by those who had decision-making power in the most important social, political, and moral issues of his era.

Senior's Theoretical Methodology

Senior believed that so much controversy existed in theories of political economy because economists had concerned themselves with social welfare rather than merely analyzing wealth. When one considered social welfare, one was immediately involved in normative or ethical statements reflecting the positions of varying contending groups involved in social conflicts. It was thus inevitable that intellectual conflicts would arise. Ethical statements, Senior declared, were not subject to either scientific confirmation or disproof. Therefore, as long as they remained a part of economic theorizing, scientific advancement could never result in agreement among theoreticians. If political economy was to become a science, it was first necessary to eliminate all of the unscientific, ethical premises contained within it. After these were eliminated, a few clearly established empirical principles of economic life would remain. Then, using deductive logic, economists would be able to explore scientifically all of the theoretical and practical implications of these few empirically substantiated principles.

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Fedric Bastiat

Scope of Fedric Bastiat utilitarian Economics

Claude Frédéric Bastiat (1801 — 1850) is one of the greatest economists ever. His role as organizer of the French, and inspiration of the nineteenth- century continental European free-trade movement is not controversial, and all historians recognize him as a great pamphleteer — some even calling him "the most brilliant economic journalist who ever lived."

Books of Bastiat:

1. *Harmonies of political economy*

In the book, Bastiat writes of "the measureless disproportion" between what each of us contributes to society and what each of us receives in return. The American economist Amasa Walker commented that, "Of all the writers on the subject, no one seems to have been more full and clear in the definition and illustration of value" than Bastiat, in the distinction he draws in *Economic Harmonies* between value and utility. Bastiat dedicated *Harmonies of Political Economy* to the youth of France.

2. *The Law*

The Law (French: *La Loi*) is an 1850 book by Frédéric Bastiat. It was written at Mugron two years after the third French Revolution and a few months before his death of tuberculosis at age 49. the work for which Bastiat is most famous, followed by the candlemaker's petition and the parable of the broken window.

Harmony and equilibrium:

Exchange is based precisely on the diversity, or, if you prefer, on the inequalities of fertility, skill, climate, and temperature, that you are seeking to eliminate.... At the race track, if one of the horses wins the prize, the other loses it; but when two horses work to produce something useful, each will produce an amount in proportion to his strength; and although the stronger will render the greater service, it does not follow that the weaker will render none at all.

Bastiat's case for the free market was however not only based on refutations of individual government interventions, but also on a general refutation of the common premise of all proposals for government action. He brilliantly argued that all these proposals involved some variant of the claim that the free market by its very nature antagonizes the interests of certain individuals or groups. This is the great subject of his unfinished *Economic Harmonies*. In the preface, which he addressed to the Youth of France, Bastiat (1851, p. 3; my translation) stressed that

What sets the various socialist schools... radically apart from the school of the economists is not this or that question of detail...; it is the point of departure, it is this preliminary and paramount question: Are human interests, left to themselves, harmonious or antagonistic?

A brief look at the history of twentieth-century economic thought confirms Bastiat's insight into the common denominator of interventionist schemes. The most important economic justifications for government intervention stressed the existence of business cycles, monopoly, public goods, and unemployment. In each of these cases, the perceived problem was claimed to be a market failure, which is nothing else but the claim that the problem under consideration springs from the very nature of the market. The market cannot possibly solve it, at any rate, it cannot solve it as well as the government, the great *deus ex machina*, which is therefore necessary to bring relief.

True to Bastiat's spirit, many Austrian and a few mainstream economists have again and again rebutted these twentieth-century allegations of market failure case by case. These rebuttals would certainly gain in strength if they were combined with a more general attack on the fallacy underlying all these individual cases. And when it comes to such endeavors, present-day economists will benefit very much from the careful study of Bastiat's doctrine of economic harmonies.

It is noteworthy that Bastiat's economic-harmonies argument contrasts sharply with the typical twentieth-century argument for the free market, inspired by Léon Walras's work, according to which the market would reach or tend to reach equilibrium, or that it would maximize or tend to maximize social utility. Thus consider how, some decades later, Louis Rougier (1938, p. 70; my translation) puts the scientific case for the free market:

Economic science demonstrates that the maximum satisfaction for trading partners is realized in a regime of free competition. In the exchange of one single product, the free play of demand and supply alone permits to derive the equilibrium price from the great number of individual bargains — the price for which the quantity of a product offered on a market will be best sold in its entirety, given a certain distribution of purchasing power among the members of society at a given point of time.

Utility and exchange :

Bastiat recognized utility as the driving force behind human action in the economic realm. Utility refers to the satisfaction or benefit that individuals derive from consuming goods and services. Bastiat argued that people act to maximize their utility, seeking to satisfy their wants

and needs. This concept of utility is foundational to understanding why individuals engage in economic activities such as production, consumption, and exchange. Exchange: Bastiat emphasized the importance of exchange in facilitating the satisfaction of human desires and needs. He viewed exchange as a voluntary transaction between individuals, where each party expects to benefit. Bastiat believed that exchange occurs because individuals have different skills, resources, and preferences, creating opportunities for mutually beneficial trade. He argued that exchange allows people to specialize in what they do best and to access a wider range of goods and services than they could produce on their own. Mutual Benefit: Bastiat contended that voluntary exchange leads to mutual benefit and overall societal welfare. When individuals freely exchange goods and services, both parties involved in the transaction are better off than they were before. Bastiat's emphasis on mutual benefit underscored his support for free trade and opposition to government intervention in economic affairs. He believed that individuals, pursuing their self-interest through voluntary exchange, contribute to the prosperity of society as a whole. Government Intervention: Bastiat criticized government intervention in the economy, particularly measures such as tariffs, subsidies, and regulations that restrict free exchange. He argued that such interventions distort the natural workings of the market, leading to inefficiency, waste, and reduced welfare. Bastiat's advocacy for free trade and limited government intervention reflected his belief in the power of voluntary exchange to promote economic progress and improve living standards.

Opposition to Protectionism

He argued that protectionist policies, such as tariffs and trade barriers, distorted the natural flow of trade and hindered economic prosperity. Bastiat famously explained the concept of opportunity cost through his "broken window fallacy," illustrating that protectionist measures might protect certain industries in the short term but ultimately lead to inefficiency, higher prices for consumers, and overall economic stagnation. He advocated for free trade as a means to promote specialization, competition, and innovation, which he believed would ultimately benefit society as a whole.

Frederic Bastiat's defense of private property, capital, profits, and rent

1. Private property :

Bastiat argued that private property is essential for human flourishing and economic progress. He believed that individuals have a natural right to own and control property, and that respect for private property rights is necessary to prevent conflict and promote cooperation in society. He saw private property as a fundamental aspect of personal freedom and individual responsibility.

2. Capital :

He recognized the importance of capital accumulation for economic growth and prosperity. He emphasized the role of savings and investment in increasing the productivity of labor and expanding the overall wealth of society. Bastiat advocated for policies that encourage capital formation, such as low taxes, minimal government regulation, and secure property rights.

Profits: Bastiat viewed profits as a legitimate reward for entrepreneurship, risk-taking, and innovation. He argued that profits serve as signals in the market, indicating where resources are most valued and where opportunities for improvement exist. Bastiat rejected the notion that profits are inherently exploitative, instead seeing them as a vital incentive for individuals to engage in productive activities that benefit society as a whole

3. Rent:

Bastiat recognized rent as the payment for the use of land or other natural resources. He believed that rent should be determined by voluntary exchange in a competitive market, reflecting the relative scarcity and desirability of different locations or resources. Bastiat opposed government intervention in rent-setting, arguing that such interference distorts market signals and hampers economic efficiency.

Role of government :

classical liberal thinker Bastiat believed that the role of government should be limited to protecting individuals' life, liberty, and property. He argued that government intervention beyond this scope often leads to unintended consequences and distortions in the economy. Bastiat famously articulated this perspective in his essay "The Law," where he criticized government policies that violated these principles, such as protectionism and wealth redistribution. He advocated for a free-market economy where individuals are free to pursue their own interests without interference from the state.

William Stanley Jevons

JEVONS' THEORY OF MARGINAL UTILITY AND EXCHANGE

William Stanley Jevons, a prominent economist from the 19th century, made significant contributions to economic theory, particularly through his magnum opus, the Theory of Political Economy, published in 1871. Within this seminal work, Jevons set the stage for a new way of looking at utilitarianism in economics. He focused on the idea of "marginal utility" and showed how this concept helps us to understand economic behavior and exchange.

Utilitarian Foundation:

Jevons strongly supported utilitarianism as the core of scientific economic theory. He believed that the value of things depended entirely on how useful they were. He emphasized the importance of thinking about the economy in terms of people's pleasure and pain, taking inspiration from Bentham's thoughts.

Exchange Value:

The ultimate truth for Jevons' was that value depends entirely upon utility. Jevons defined value exclusively in terms of 'exchange value' or 'price', rejecting labor theories of value. He directed his attention solely towards prices, advocating for a market-centric economic analysis.

Economic Agents:

People, in Jevon's view, had only 2 characteristics that defined them as economic agents:

1. The first characteristic was that they derived utility from consuming commodities.
2. The second characteristic was that every person was a rational, calculating maximizer. And rational, calculating, maximizing behavior was the only element of human action to be studied by economics.

Jevons postulated that individuals are distinguished by their utility-maximizing behavior and derive satisfaction from consuming commodities. Rational, calculating maximization governs economic decision-making.

Marginal Utility:

Jevons introduced the concept of marginal utility, distinguishing between total utility (derived from the consumption of a given quantity of some commodity) and the 'final degree of utility' (derived from consuming the last increment of a commodity).

Although total utility might continue to become greater as one consumes an increased quantity of a commodity, the final degree of utility ultimately decreases as that quantity increases. It was this "final degrees of utility" or marginal utility, that concerned Jevons.

The principle of diminishing marginal utility became foundational in neoclassical economics.

Mathematical Exposition:

By introducing the concept of "marginalism" to utilitarian economics, Jevons discovered a method to translate the idea that people act rationally to maximize their gains into mathematical terms.

If the total utility one received consuming a commodity depended on the quantity consumed, then this could be written as a mathematical function, $TU = f(Q)$. Here, TU denotes total utility and Q represents the quantity consumed.

In calculus, the first derivative of a function tells one how much the dependent variable (In this case, total utility) changes as a consequence of an infinitesimally small change in the independent variable (in this case, the quantity consumed).

The first derivative of the total utility function gives one the marginal utility at any particular quantity consumed.

Jevons employed calculus to formalize the logic of maximization, explaining how marginal utility determines optimal consumption levels. Total utility function was maximized when the quantity was increased to that point at which marginal utility was equal to zero. It simply meant that to maximize one's utility from consuming a particular commodity, one should consume the commodity until one was satiated, that is, until one could derive no more utility from another small increment of the commodity.

Equilibrium in Exchange:

When consumption involves cost, those costs can be stated in mathematical form. For example, if one possessed a commodity 'y' and could only get another commodity 'x' by giving up some

'y' in exchange, then one could compare the reaction of one's marginal utilities for the two commodities, (MU_x / MU_y) and the prices of two commodities, (P_x / P_y) .

If $(MU_x / MU_y) > (P_x / P_y)$, then one could gain utility by trading some of his 'y' for some 'x'. The process would continue until the Individual exhausted the gain from exchange, he would have traded to the point at which $(MU_x / MU_y) = (P_x / P_y)$.

To put the same thing differently, the ratio (MU_x / P_x) would tell one how much additional utility one could get or give up, if one purchased or sold an additional dollar's worth of commodity 'x'. The two individuals would, Jevons asserted, buy and sell the commodities with the marginal utilities of each commodity, changed to the point where $(MU_x / P_x) = (MU_y / P_y)$.

At that point, the last dollar's worth of either 'x' or 'y' yielded the same increment to the individual's total utility. This is referred as the 'Equimarginal Principle'

William Stanley Jevons' Theory of Marginal Utility and Exchange changed our thoughts about economics by combining utilitarian principles with mathematical analysis. His emphasis on marginal utility, rational behavior, and market equilibrium set the stage for modern neoclassical economics, shaping our comprehension of consumer choice, exchange dynamics, and resource allocation. Jevons' work remains a key part of economic theory, influencing both academic study and policy-making.

Carl Menger

Menger's Theory of Marginal Utility, Prices & Income Distribution

Carl Menger was born on 23 February, 1840 in Neu-Sandec, Galicia (now Poland). He was a prominent Austrian economist. He is renowned for developing the marginal utility theory. Menger received his Ph.D from Jagiellonian University in 1867 and later became a professor of Political Economy at University of Vienna. Among Menger's most important discoveries are the solutions to the diamond-water paradox, which show that products have value because of their multiple, varying applications, and the debunking of the labor theory of value, which shows that goods' worth comes from meeting human needs rather than labor input.

The book "Principles of Economics," written by Carl Menger and published in 1871, is his most famous contribution. It provided noble concepts that contradicted traditional economic theories and served as the basis for the Austrian School of economics. He also made important contributions in methodology, some of them are included in the English translation titled 'Problems of Economics and Sociology'. In his economic theory, he rejected the use of mathematical equations and expressed his theories verbally through numerical examples. Although there are similarities with Jevons theory, he illustrated his marginal utility principle with a table of number that we can see below:

Number of units consumed	Type of commodity									
	I	II	III	IV	V	VI	VII	VIII	IX	X
1	10	9	8	7	6	5	4	3	2	1
2	9	8	7	6	5	4	3	2	1	0
3	8	7	6	5	4	3	2	1	0	
4	7	6	5	4	3	2	1	0		
5	6	5	4	3	2	1	0			
6	5	4	3	2	1	0				
7	4	3	2	1	0					
8	3	2	1	0						
9	2	1	0							
10	1	0								
11	0									

Table: An Illustration of Declining Marginal Utility

Table (1) reproduces the numbers from Menger's table with some labels added for clarification. In the table, in order to get marginal utility of some commodity, say commodity III, one needs to

go through the third column and then read down to the number of units consumed. If 7 units of commodity III is consumed, the marginal utility of the seventh unit is 2. However, total utility is simply the summation of all the marginal utilities down to the number of units consumed. For this example, total utility is 35. so , seven units of commodity III yields a total utility of 35.

The relationship can also be represented through graphs. There in the figure 1 we can clearly see the relationship between the marginal utility and the total utility of commodity III. This relationship between Marginal and total utility is a common phenomenon in neoclassical economics and we can see its recurrences in many cases. We can also analyze several types of maximizing problems in such manners. Here, in the graph we can clearly see that, for consuming one unit of commodity III, total and marginal utility is 8 units, in the case of 7th unit marginal utility is 2 but total utility is 35. Lastly, the 8th unit generates 1 unit of marginal utility and 36 units of total utility. It is noticeable that our utility remains 36 as marginal utility is 0

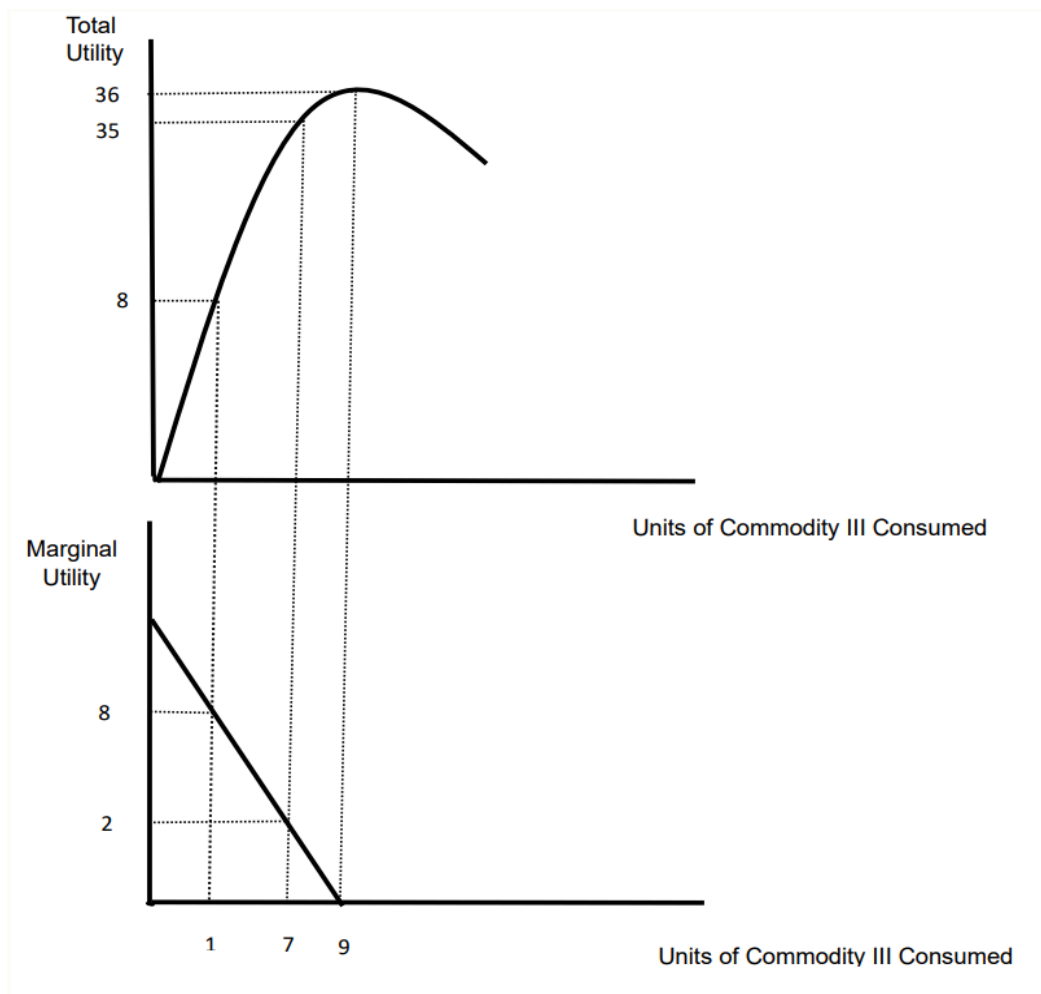


Figure-1: Relationship Between Total Utility & Marginal Utility

and any further consumption will cause negative marginal utility and total utility will fall. That's the reason after the 8th unit our total utility curve is moving downward.

According to Menger, when a person equals the marginal utility obtained from any one commodity to the marginal utility acquired from each other commodity they consume, they have reached the equilibrium at which they maximize their utility. This statement of maximizing condition was not as good as Jevon's because Menger ignored pricing. Only if the price per unit of each of Menger's commodity categories was equal to the price per unit of every other type would his maximizing solution remain true. In that case, Jevon's equation $MU_x / P_x = MU_y / P_y$ would be the same as Menger's. Both Jevon's and Menger's formulations were accurate, though Menger was unaware that his formulation was unique, extremely unlikely cases.

Menger's explanation of how prices are determined was better than Jevon's. Menger used supply and demand to explain pricing. Every classical economist had firsthand experience with supply and demand-driven short-term market prices. Thus, there has never been a difference between classical, Marxist, and neoclassical economists in this regard. Their explanations of the principles underlying supply and demand have differed. For example, Marx, Ricardo, and Smith all attempted to explain profit, wage, and rent. In reality, these belonged outside the domain of prices since they affected both the cost of an individual firm's output and the distribution of income among social classes. For cost components they used Smith's 'natural price' or Marx's 'price of production' of a commodity. In equilibrium market price which is determined by supply and demand will be equal to natural price or price of production. This pricing process is different from Menger's. The pricing process is from the labor theory perspective. The utility perspective showed the pricing process in a completely different manner. Here, Menger explained supply and demand through utility. As a result, utility ultimately determined the prices of consumer goods as well as the prices of the factors of production (land, labor, capital). Their owners' (factors of production) utility estimates set their supply, while their productivity in creating consumer goods and the utility that consumers receive from using them drove their demand. As a result, in this perspective, wages, rent and profits were at least partially determined by the prices of consumer goods. There was a noticeable distinction between the two perspectives on the pricing procedure. Menger outperformed Jevon in this pricing procedure because of the way he used his utility theory framework to explain it.

Menger provided an explanation for the demand for consumer goods by demonstrating that most consumers would obtain greater utility by holding onto their money rather than spending it on a particular good when that good's price was high relative to the marginal utility that most consumers could obtain from it. As the cost of the thing decreased, however, a greater number of consumers discovered that the marginal utility they gained by using it outweighed the benefit

they lost by forgoing the lesser dollar amount. Additionally, when the price decreased, those who were already consuming the commodity would discover that purchasing more of it was necessary to maximize its utility. Therefore the law of demand was derived from Menger's theory of diminishing marginal utility: "The quantity of a commodity that people were willing to purchase depended on the price of the commodity, and the quantity demanded and the price are inversely related to each other." However, Menger's supply explanation was less adequate. Typically, he viewed supply as an amount that was already in the seller's possession. The vendor chose the quantity he or she wanted to sell at each price by applying utility maximization as a guide. Prices were set by the sum of the purchase and sell wants, which were all influenced by utility considerations.

Menger divided all sorts of goods into two categories:

1. "Goods of first order" which refers to the commodities produced for consumption. and,
2. "Good of higher order" which means the factors of production.

He tried to establish a relationship between these two types of goods .First, in order to manufacture "goods of lower order," a range of higher order goods were required; as a result, capital and land were just as important to production as labor. He agreed with the labor theory of value in this remark. Secondly, he aimed to demonstrate that "the value of goods of higher order is always and without exception determined by the prospective value of the goods of lower order in whose production they serve," which is a clear contradiction to the labor theory. Regarding this, he thought that "among the most egregious of the fundamental errors that have had the most far-reaching consequences in the previous development of our science" was the labor theory of value.

Menger's Argument on Methodology:

His initial claim was that "pure science" was invariably devoid of value. He thought that normative, moral, or ethical principles had no place in science. People's values were shaped by their experiences, emotions, and social class; as a result, there was never much chance of broad consensus on moral issues. But science was the study and depiction of reality as it is, not as one would like it to be. Which means a science economist should be value free and he believed that ethical judgements had no place in pure economic theory. He wrote the so-called "ethical orientation of political economy" was vague and confusing.

Menger's second methodological tenet was that only individual families or business firms ,what he dubbed "individual economies" could be scientifically understood by economists. They were never able to arrive at a comparable scientific comprehension of social aggregates like nations or classes. Menger was not a fan of the term "national interest," which was so often invoked to support calls for national reform. He thought it was based on the fallacious idea that one could

identify and examine a national economy with its own interests, independent of "individual economies" and their interests.

Leon Walras

Walras's Theory of General Economic Equilibrium

Similar to Jevons and Menger, Walras independently developed the theory of diminishing marginal utility and discussed the equation that gave the maximum utility a consumer could achieve through exchange. Walras's most important and lasting contribution to economic theory was his theory of general economic equilibrium.

Although most previous economic theorists have discussed interrelationships between different markets, no economists prior to Walras set out a general conceptual and theoretical structure within which to examine the multiplicity of relationships among different markets. He realized that the forces of supply and demand in any one particular market depended in varying degrees on the prices prevailing in innumerable other markets.

In a general theory of price determination, all prices would have to be determined simultaneously by both the total of all consumer utilities and the interrelations that existed among all markets. Such interrelationships did not exist only in the demand for consumer goods. They also existed in the supply of consumer goods, as well as in the demands and supplies of other types of commodities or assets that were exchanged. Walras attempted to formulate a general theoretical framework within which he could show how, through the interactions of all markets, all prices could be determined simultaneously.

The institutional setting of Walras's theory was competitive capitalism in which "there are land-owners, workers and capitalists." These three classes functioned in two economically important ways:

- i. First, as owners of productive services (land, labor, and capital), they supplied these services in the market.
- ii. Second, as consumers, they demanded consumer goods in the market.

Taking the existing distribution of “productive services” and different people's various roles in the production process, the most important element within the socioeconomic context was, for Walras, people’s “subjective desires” or their schedules of marginal utilities.

There were three important institutional factors in Walras’s theory:

- i. The acceptance of the existing laws of and distribution of property as morally right and just. Here he simply stated that
“The ownership of property... is legalized appropriation, or appropriation in conformity with justice.”
- ii. He assumed that the economy was made up exclusively of small, relatively powerless business firms, and perfect competition prevailed. Although he realized that “the principle of free competition is not generally applicable...[to] monopolies”, he ignored monopolies entirely throughout his general equilibrium discussion.
- iii. People have simply assumed to have measurable marginal utility schedules. There was no discussion of how their utility schedules came into existence or how they changed overtime. He simply assumed that during the time period under analysis “the utility... remains fixed for such party”, Thus utility was a metaphysically given, ultimate datum, in terms of which everything was to be explained.

After assuming the existence of an existing distribution of ownership, a perfectly competitive market and metaphysically given, fixed utility schedules, Walras developed his system of equations to show what quantities of the various commodities were exchanged and how their prices were determined.

Walras’s law is really a definitional identity which states that with any given set up prices, the total demand for all things must equal the total supply of all things exchanged. Therefore, every individual demand is simultaneously a supply of the same magnitude, and so if these individual demands and supplies are aggregated, the totals must be equal.

After Walras, Kenith Arrow and Keynes discussed general equilibrium.

END

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